

TRANSMISSION LINES & WAVEGUIDES (ECA15)

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EXP NO.2

Analysis of SWR (Standing Wave Ratio) in Transmission Lines by Varying Load Impedance and Characteristic Impedance

AIM

To analyze the standing wave ratio (SWR) by varying load impedance (ZL) for a fixed characteristic impedance (Z0), and by varying characteristic impedance (Z0) for a fixed load impedance (ZL).

APPARATUS / SOFTWARE REQUIRED

SCILAB Version 6.1.1

PROCEDURE

- Open SCILAB software.
- Open File → New and create a new program.
- Enter the required impedance values.
- Type and execute the program.
- Record the SWR values displayed in the console.
- Repeat for the second test case.
- Compare simulation results with theoretical SWR values.

CALCULATION

Reflection coefficient: $K = (Z_L - Z_0) / (Z_L + Z_0)$

Magnitude: $|K| = \sqrt{[\text{Re}(K)]^2 + [\text{Im}(K)]^2}$

Standing wave ratio: $\text{SWR} = (1 + |K|) / (1 - |K|)$

TEST CASE 1: VARYING LOAD IMPEDANCE

Fixed characteristic impedance $Z_0 = 50 \Omega$; load impedance values are 25Ω , 50Ω , 75Ω and 100Ω .

CODE FOR VARYING LOAD IMPEDANCE :

```
clc;

clear;

clc;

Zo = 50;

ZR_values = [25 50 75 100];

disp("TEST CASE 1: VARYING LOAD IMPEDANCE");

for k = 1:4

    ZR = ZR_values(k);

    K = (ZR-Zo)/(ZR+Zo);

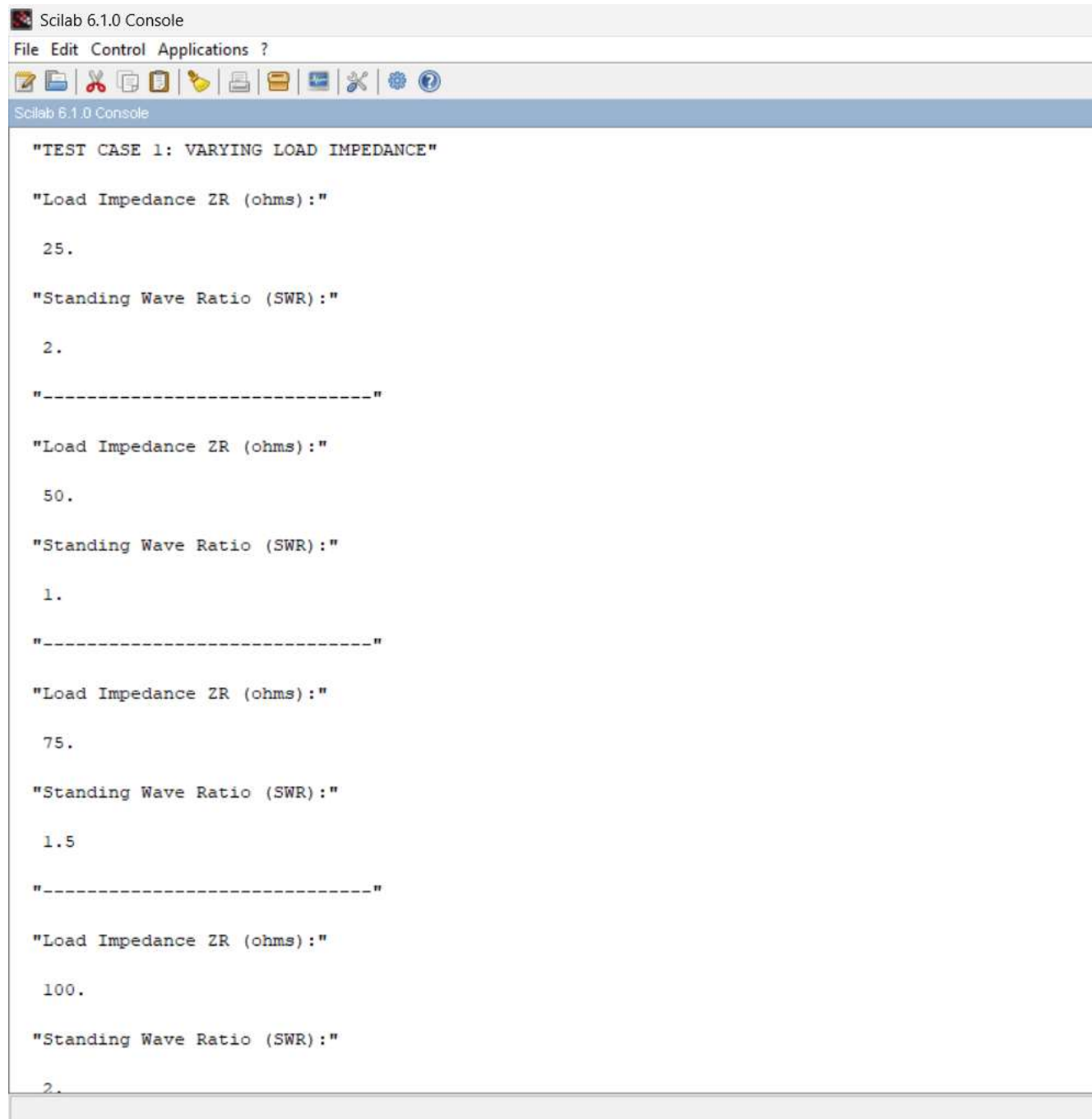
    ampK = abs(K);
```

```

S = (1+ampK)/(1-ampK);
disp("Load Impedance ZR (ohms):");
disp(ZR);
disp("Standing Wave Ratio (SWR):");
disp(round(S*1000)/1000);
disp("-----");
end

```

OUTPUT SCREENSHOT :



```

Scilab 6.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 6.1.0 Console

"TEST CASE 1: VARYING LOAD IMPEDANCE"

"Load Impedance ZR (ohms):"

25.

"Standing Wave Ratio (SWR):"

2.

"-----"

"Load Impedance ZR (ohms):"

50.

"Standing Wave Ratio (SWR):"

1.

"-----"

"Load Impedance ZR (ohms):"

75.

"Standing Wave Ratio (SWR):"

1.5

"-----"

"Load Impedance ZR (ohms):"

100.

"Standing Wave Ratio (SWR):"

2.

```

TABLE

ZL (Ω)	Simulation SWR	Theoretical SWR
25	2.000	2.000
50	1.000	1.000
75	1.500	1.500
100	2.000	2.000

RESULT – TEST CASE 1

The SWR was calculated for a fixed characteristic impedance of 50 Ω while varying load impedance. When $Z_L = Z_0$, SWR = 1, indicating a matched condition with no reflection.

TEST CASE 2: VARYING CHARACTERISTIC IMPEDANCE

Fixed load impedance $Z_L = 50 \Omega$; characteristic impedance varies from 250 Ω to 400 Ω as shown in the supplied console outputs.

CODE FOR VARYING CHARACTERISTIC IMPEDANCE :

```
clc;

clear;

// Fixed load impedance

ZL = 50;

// Characteristic impedance values

Zo_values = [250 275 300 325 350 375 400];

disp("TEST CASE 2: VARYING CHARACTERISTIC IMPEDANCE");

for k = 1:7

    Zo = Zo_values(k);

    K = (ZL - Zo)/(ZL + Zo);

    ampK = abs(K);

    S = (1 + ampK)/(1 - ampK);

    disp("Characteristic Impedance Zo (ohms):");

    disp(Zo);

    disp("Standing Wave Ratio (SWR):");

    disp(round(S*1000)/1000);

    disp("-----");

end
```

OUTPUT SCREENSHOT

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

"TEST CASE 2: VARYING CHARACTERISTIC IMPEDANCE"

"Characteristic Impedance Zo (ohms):"

250.

"Standing Wave Ratio (SWR):"

5.
"-----"
"Characteristic Impedance Zo (ohms):"

275.

"Standing Wave Ratio (SWR):"

5.5
"-----"
"Characteristic Impedance Zo (ohms):"

300.

"Standing Wave Ratio (SWR):"

6.
"-----"
"Characteristic Impedance Zo (ohms):"

325.

"Standing Wave Ratio (SWR):"
```

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

325.

"Standing Wave Ratio (SWR):"

6.5
"-----"
"Characteristic Impedance Zo (ohms):"

350.

"Standing Wave Ratio (SWR):"

7.
"-----"
"Characteristic Impedance Zo (ohms):"

375.

"Standing Wave Ratio (SWR):"

7.5
"-----"
"Characteristic Impedance Zo (ohms):"

400.

"Standing Wave Ratio (SWR):"

8.
"-----"

-->
```

TABLE

Z0 (Ω)	Simulation SWR	Theoretical SWR
250	5.000	5.000
Z0 (Ω)	Simulation SWR	Theoretical SWR
275	5.500	5.500
300	6.000	6.000
325	6.500	6.500
350	7.000	7.000
375	7.500	7.500
400	8.000	8.000

RESULT – TEST CASE 2

The SWR was calculated for a fixed load impedance of 50 Ω while varying characteristic impedance from 250 Ω to 400 Ω . SWR increases as the impedance mismatch increases.

OVERALL RESULT

The standing wave ratio was analyzed using SCILAB for two cases: varying load impedance with fixed characteristic impedance, and varying characteristic impedance with fixed load impedance. SWR equals 1 when $Z_L = Z_0$ and increases as the impedance mismatch increases.